Biology :-

MCQs :-

1. D
2. B
3. A
4. D
5. D
6. B
7. A
8. C
9. C
10. B
11. C
12. A.

Q:2(i)(a)

A: 2(i)(a)

Muslim scientists :-

① Jabir bin Hayyan
② Abdul malik Asmai
③ Bu ali Sina

Biology is ~~dir~~ the main division of three fields:-

① Botany : The study of plants.
② Zoology :- The study of animals.
③ Microbiology :- The study of microorganisms.

Q:2(ii)(or)

A:2(ii)(or)

① Autotrophs :-
Are organisms that are capable of producing their own food. (photosynthetic mode of nutrition etc).

② Heterotrophic :-
Are organism that are capable of eating other things as their food (~~absorptive~~ ingestive mode of nutrition).

③ Saprotrophs :-
Are organisms that depends on dead, decaying matter (absorptive mode of nutrition)

Q:2(iii)(or)

A:2(iii)(or)

Taxa	Pea
Kingdom	Plantae
Division / Phylum	Magnoliophyta
Class	Magniopsida
Order	Fabales
Family	Fabaceae
Genus	Pisum
Species	Pisum sativum.

Q:2(iv)

A:2(iv)

Middle lamella :-
Made up of Mg and Ca salts of pectin, it is sticky in nature, holds the neighbouring cell walls together.

Primary wall :-
Contains cellulose fibres arranged in a criss cross fashion, it is thin and flexible.

Secondary wall :-
formed inside the primary cell wall in some cells. It is a rigid due to the presence of lignin.

Plasmodesmata :-
Tiny pores in the cell wall through which ~~~~ neighbouring cells form cytoplasmic connections.

Q:2(v)(or)

A:2(v)(or)

Ribosomes:-
Are composed of roughly equal amounts of proteins and rRNA, made up

of 2 subunits, large and small subunits which join together to perform their functions.

They are tiny granular structure, not bound by any membrane, found in both prokaryotic and eukaryotic ~~cells~~ proteins made up of about 55% of the dry weight of cell, much of which is synthesized by ribosomes.
Prokaryotic ribosomes are smaller in size than eukaryotic ribosomes.

Q: 2(vi) (or)
A: 2(vi) (or)

Chromatid :-
1 of the 2 identical copies of chromosome, formed after DNA replication.

Centromere :- The point at which two chromatids are attached to eachother.

During S-phase, the cell replicates it's genetic information to form the exact two copies, that is why chromosomes have 2 chromatids (sister), one for each of the daughter cell.

Q:2(vii)

A:2 (vii)

Lymphatic system:-
Organs / structure are lymph nodes, lymph, lymph vessels role- defends against infection or disease and maintains tissue fluid homostatis.
Immune system:- Organs are Leukocytes, ~~Pathogens~~ : tonsils, adenoids, thymus, and spleen, roles - removes pathogens fights infections and helps in healings.

Q:2 (viii)(or)

A:2 (viii)(or)

Cardiovascular system:-
Heart, blood and blood vessels helps stablize levels of gases, wastes, body temperature and pH.

Skeltal system:-
Bones, cartilage, joints, tendons, and ligaments; regulates the level of calcium and other minerals in the ~~body~~ blood by storing or releasing them from bones

Q:2(ix)
A:2(ix)

Nervous System:-
Brain, spinal cord, nerves and sensory organs (eyes, ears, tongue, skin, nose); maintains homeostasis by rapidly controlling body functions and directing short-term change in other systems.

Respiratory system:-
Mouth, ~~Rhp~~ Pharynx, larynx, tranchea, bronchi, lungs and diapharynx, controls the balance of oxygen and CO_2 in the body.

Q:2(x)(or)
A:2(x)(or)

Potassium
Sulphur
Chlorine
~~Ca~~ Sodium
Magnessium
Iron
Copper
Maganesse
Zinc
Iodine

Q:2(xi)
A:2 (xi)

Many enzymes work together in a specific sequence to make metabolic pathways. A metabolic pathway is a series of connected chemical reactions.

Long questions.

Q:3(i)
A:3(i)

Histology :-
The ~~study~~ microscopic study of tissues of organism.

Genetics :-
The study of genes and heridity in organisms.

Embroyology :-
The study of developement of an organism form a fertilized egg.

Paleontology :-
The study of the histry of the life on Earth based on

fossil fuels.

Immunology:- The study of the immunology.
The ability of the body to protect itself from substances like microbes.

Q:3(ii)
A:3(ii)

Plant cell	Animal cell
1 Cell wall cements adjoining cells, forming supportive structure.	Support depends on collective arrangement of tissues organs and skeletal system (no cell wall needed).
2 Transports channels, xylem and phloem are formed bcuz of cell wall	Transport of fluids, ~~nutot~~ nutrients, and gases occurs through different structure and mechanisms.
3 Rigid cell wall helps withstand high osmotic stress and store water.	Cannot withstand high osmotic pressure or store large volume of water.

(4) Can become turgid, which keeps the plant upright.	Cannot become turgid.
(5) Cannot move from place to place due to rigidity of the cell wall.	Lacking of cell wall animal cell are flenible and can move to find shelter, food or better reproductive opportunities.

Q.3(iii)

A: 3(iii)

Mitosis	Meosis
1. Occurs in animal cell	Occurs in germ line cells.
2. Cell divides in only once.	Cell divides twice.
3. Produces 2 daughter cell.	Produce 4 daughter cell.

i) Daughter cell produced are diploid.	All daughter cell produced are haploid
i) Daughter cells prod become part of somatic body.	Daughter cell form gametes.
i) All chromosomes remain independent of each other.	Homologous chromosomes pair with each other. (synapsis).
i) No chaismata formation	Chaismata are formed
i) No crossing over occurs during the process.	Crossing over takes place during meosis I.

Q:3(iv)

A:3 (iv)

Lipids are mainly made up of fatty acids and glycerol (an alcohol), collectively called acylglycerids. The most common type of acylglycerides is the tri-acyl glyceride, which contains 1 glycerol and 3 fatty acids (glycerols + 3 fatty acids → tri acylglycerides)

Saturated fatty acid:-
have all C-C single bond and are solid at room temperature e.g cheese, butter, ghee

Unsaturated fatty acids:-
have double or triple C=C bonds and are liquid at room temperature e.g soyabean oil, mustard oil, olive oil.

Lipids functions:-

① Phospho lipids and cholestoral are components of the plasma membrane structure.

② Lipids act as energy stores in fat cells, the livers and the blood; 1 grams of lipid provides 9.1 Kcal of energy, double the energy ~~food~~ provided by carbohydrates or proteins.